Automobile Sales & Service Data Analytics Project

Data Visualization using Power BI

Personal Project

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Tools: Power BI | Excel | DAX

May 2026

1. Project Overview

Industry: Automobile

Domain: Sales, Marketing Performance and After-Sales Service & Maintenance

Background

The automobile industry is going through a pretty significant shift right now. Between the growing push for electric vehicles, advances in self-driving tech, and changing customer expectations around digital experiences, businesses in this space have a lot to keep up with. In this kind of environment, having a clear picture of both sales performance and after-sales service isn't just helpful — it's essential for staying competitive and profitable.

On the sales and marketing side, the competition is tough. Companies need smart strategies to bring in new customers, convert leads, and forecast demand accurately. Questions like how to spend the marketing budget wisely, which customer segments to target, and how individual dealerships are performing are all really important. Data analytics helps cut through the noise and find answers — like which models are selling well, which marketing channels are actually working, and where sales are strongest by region.

After-sales service is just as important, though it sometimes gets overlooked. Good service keeps customers coming back, builds brand loyalty, and is a serious revenue stream on its own. The challenge is staying efficient — handling service appointments well, managing parts inventory, and keeping customers happy. And with newer, more complex vehicles (especially EVs), service centers are having to constantly adapt. This project brings both sides together to give a complete view of the business, so decisions can be made based on actual data rather than guesswork.

2. Problem Statements / Key Business Questions

The whole point of this project was to use data analytics to answer real questions that a business in the automotive space would actually care about. Here's what I set out to find:

Sales & Marketing Questions:

- What do monthly, quarterly, and annual sales look like when broken down by vehicle model, type, and region?
- Which models are top sellers and which ones aren't pulling their weight in terms of units sold and revenue?
- How is actual sales revenue stacking up against the targets set for each dealership?
- What's the average selling price per model, and does it vary across dealerships or regions?
- Which marketing channels — online ads, dealership visits, referrals — are actually driving sales?
- What demographic profile do buyers of different vehicle types tend to have?

After-Sales Service Questions:

- What percentage of service visits get resolved on the first try, and how does that vary by service type or technician?
- What are the most common issues customers come in for, and which models tend to have the most problems?
- How long does it typically take from drop-off to delivery, and where can the process be sped up?
- How much revenue is the service department generating, and how is it split across service types and dealerships?
- What are customers actually saying about their service experience, and how does satisfaction change over time?
- Are customers who had a good buying experience more likely to come back for service?
- Which parts get replaced most often, and what do they contribute to overall service costs?

3. Data Design

Good data structure is what makes any analytics project actually work. For this one, I went with a star schema — it keeps things organized by separating the actual transaction data (fact tables) from the descriptive information (dimension tables). This makes Power BI run faster and makes it much easier to build meaningful visuals.

Sales Fact Table — Fields:

- SaleID
- SaleDate
- VehicleID
- CustomerID
- DealershipID
- SalePrice
- DiscountAmount
- FinancingType
- MarketingChannel
- SalespersonID
- TargetRevenue

Service Fact Table — Fields:

- ServiceOrderID
- ServiceDate
- VehicleID
- CustomerID
- DealershipID
- ServiceType
- ServiceCost
- PartsCost
- LaborHours
- ServiceStatus
- FirstTimeFix
- CustomerSatisfactionScore
- TechnicianID

Dimension Tables:

Customer Master: CustomerID, CustomerName, Email, Phone, Address, City, State, ZipCode, AgeGroup, IncomeBracket

Vehicle Master: VehicleID, Make, Model, Year, VehicleType, EngineType, MSRP

Dealership Master: DealershipID, DealershipName, Region, City, State, AnnualSalesTarget, AnnualServiceTarget

Data Types Used:

- **Text:** All ID fields, names, categories, and descriptive columns
- **Number:** Prices, costs, scores, quantities
- **Date:** SaleDate, ServiceDate
- **Derived/Calculated:** Any computed fields like TotalSales

4. Data Acquisition

Since real automotive sales and service data is generally private and hard to get hold of, I put together a simulated dataset that closely mirrors what you'd find in a real-world scenario. I referenced resources from GeeksforGeeks, Kaggle, and used Gemini to help shape the structure. The data was built in Microsoft Excel and Google Sheets, which gave me full control over how it was structured and what kind of data quality issues it contained.

The dataset covers roughly 8,000 sales records and 5,000 service records, spanning January 2022 to December 2023. On top of that, there are matching dimension tables for customers, vehicles, and dealerships.

5. Data Transformation (ETL)

All the cleaning and transformation work was done inside Power BI using Power Query Editor. Here's a rundown of what I did:

1. **Data Loading:** Brought all five files — Sales.csv, Service.csv, Customers.csv, Vehicles.csv, and Dealerships.csv — into Power Query Editor to get started.
2. **Fixing Data Types:** Made sure date columns were properly set as Date type, numeric fields like prices and scores were set to Decimal or Whole Number, and all ID columns were kept as Text to avoid them being accidentally added up.
3. **Handling Missing Values:** For important numeric fields like SalePrice or ServiceCost, any rows with nulls were removed since those records would be useless for financial analysis. For optional fields like DiscountAmount, I replaced nulls with 0 so the numbers still worked without losing rows. Missing contact details in the Customer table were left as-is since they don't affect the analysis.
4. **Fixing Incorrect Formats:** Tracked down any text values hiding in numeric columns (like 'N/A' in a price field) and either replaced them or removed the row. Also made sure all dates were in a consistent YYYY-MM-DD format.
5. **Removing Duplicates:** Removed duplicate rows using the primary key in each table — SaleID for Sales, ServiceOrderID for Service, and so on — to make sure nothing was being double-counted.
6. **Standardizing Column Names:** Cleaned up column names to remove extra spaces and make them consistent. This makes writing DAX formulas a lot less frustrating.
7. **Text Cleanup:** Applied Trim to all text columns to get rid of stray spaces. Also used Proper Case on descriptive fields like Make, Model, and DealershipName so visuals look clean and consistent.

6. Exploratory Data Analysis (EDA)

Before diving into building dashboards, I spent time exploring the data to understand what it looked like, catch any patterns, and spot anything unusual. This step is important because it helps you know what questions are actually answerable before you commit to a design.

Dataset Overview:

- **Sales Table:** Around 8,500 individual vehicle sale records covering transaction date, price, customer, vehicle, dealership, and marketing channel.
- **Service Table:** About 5,200 service repair orders with details on service type, cost, labor hours, and customer satisfaction.
- **Customer Master:** Close to 980 unique customer profiles with demographic info.
- **Vehicle Master:** 115 distinct vehicle models with specs like make, model, year, and type.
- **Dealership Master:** 8 dealerships with location details and annual sales/service targets.

Key Observations:

- Total sales revenue across all dealerships came to roughly $350 million, with an average of around $41,000 per vehicle — suggesting a focus on the mid-to-high end of the market.
- Sedans and SUVs made up the bulk of sales, but EV sales grew by about 20% year-over-year, which is a clear sign of shifting preferences.
- Sales were consistently highest in Q4, probably driven by seasonal promotions and year-end offers.
- The service department brought in around $3.1 million in total revenue, with an average service cost of about $596 per order.
- Scheduled maintenance was the most common type of service visit, followed by general repairs. Brake service, oil changes, and tire rotations came up most frequently.
- The overall First-Time Fix Rate (FTFR) was around 85%, which is decent but shows there's room to improve — especially for more complex repairs.
- Average Customer Satisfaction Score for service was 4.2 out of 5. Dealerships with higher FTFR also tended to score higher in customer satisfaction.

7. Data Model

The data model in Power BI is built around a star schema — two fact tables at the center (Sales and Service), connected to three shared dimension tables (Customers, Vehicles, Dealerships). All relationships are one-to-many, with dimension tables on the one side and fact tables on the many side.

Relationships Defined:

- **Customers** → **Sales** via CustomerID — One customer can have many sales transactions. Cardinality: 1:* | Filter direction: Single
- **Customers** → **Service** via CustomerID — One customer can have many service orders. Cardinality: 1:* | Filter direction: Single
- **Vehicles** → **Sales** via VehicleID — One vehicle model can appear in many sales. Cardinality: 1:* | Filter direction: Single
- **Vehicles** → **Service** via VehicleID — One vehicle model can have many service records. Cardinality: 1:* | Filter direction: Single
- **Dealerships** → **Sales** via DealershipID — One dealership can process many sales. Cardinality: 1:* | Filter direction: Single
- **Dealerships** → **Service** via DealershipID — One dealership can handle many service orders. Cardinality: 1:* | Filter direction: Single

8. KPI Development (DAX Measures)

To actually answer the business questions, I built out a set of KPIs using DAX in Power BI. These are dynamic measures, meaning they automatically update based on whatever filters or slicers are active in the report.

Sales & Marketing KPIs:

Total Sales Revenue: Adds up the total money made from all vehicle sales — the most fundamental measure of overall performance.

```
Total Sales Revenue = SUM(Sales[SalePrice])
```

Total Units Sold: Counts how many vehicles were sold in total, which helps understand volume and model popularity.

```
Total Units Sold = COUNTROWS(Sales)
```

Average Sale Price: Works out the average price per vehicle sold. Useful for analyzing pricing strategy and product mix.

```
Average Sale Price = DIVIDE([Total Sales Revenue], [Total Units Sold], 0)
```

Total Sales Target: Adds up the annual sales targets across all dealerships to give a combined goal to measure against.

```
Total Sales Target = SUM(Dealerships[AnnualSalesTarget])
```

Sales Target Achievement %: Shows how close actual revenue is to the target — great for spotting over and under-performers.

```
Sales Target Achievement % = DIVIDE([Total Sales Revenue], [Total Sales Target],
0)
```

Sales % by Channel: Shows what proportion of total sales came from each marketing channel.

```
Sales % by Channel = DIVIDE([Total Units Sold], CALCULATE([Total Units Sold],
ALL(Sales[MarketingChannel])))
```

After-Sales Service KPIs:

Total Service Revenue: Totals up all income from service and maintenance work.

```
Total Service Revenue = SUM(Service[ServiceCost])
```

Total Service Orders: Counts the number of service appointments handled.

```
Total Service Orders = COUNTROWS(Service)
```

Average Service Cost per Order: Calculates the average cost per service visit — helpful for efficiency analysis.

```
Average Service Cost per Order = DIVIDE([Total Service Revenue], [Total Service
Orders], 0)
```

First Time Fix Count: Counts how many service issues were fully resolved on the first visit.

```
First Time Fix Count = CALCULATE(COUNTROWS(Service), Service[FirstTimeFix] =
"Yes")
```

First Time Fix Rate %: The percentage of service visits resolved on the first try — one of the most important service quality metrics.

```
First Time Fix Rate % = DIVIDE([First Time Fix Count], [Total Service Orders], 0)
```

Average CSI: The average customer satisfaction score across all service visits.

```
Average CSI = AVERAGE(Service[CustomerSatisfactionScore])
```

Total Service Target: Combined annual service revenue target across all dealerships.

```
Total Service Target = SUM(Dealerships[AnnualServiceTarget])
```

Service Target Achievement %: Same idea as the sales target measure, but for the service side.

```
Service Target Achievement % = DIVIDE([Total Service Revenue], [Total Service
Target], 0)
```

9. Dashboard Insights

The final Power BI dashboard has multiple pages of interactive visuals, each focused on answering a specific set of business questions. Here's a summary of what each section shows:

Sales by Vehicle Category: SUVs and Sedans are clearly the top earners, which makes sense given how popular those categories are with buyers looking for utility and comfort.

Sales by Region and Dealership: The South and West regions consistently perform the best. Dealerships in cities like Mumbai, Bengaluru, and Chennai are the biggest contributors to overall revenue.

Customer Loyalty Analysis: Platinum and Gold tier customers are responsible for most of the high-value transactions, and they tend to come back repeatedly. This suggests loyalty programs are worth investing in.

Time Series Analysis: Sales spike during Q4 — particularly November and December — which lines up with festive season offers and year-end promotions.

Top Vehicles and Customers: A handful of models dominate the charts, and a relatively small group of loyal customers account for a large chunk of repeat business.

Visual Components Used:

- KPI Cards for quick-glance metrics
- Line charts for monthly and quarterly sales trends
- Bar charts for top-selling vehicles and dealership comparisons
- Map visual for regional sales distribution
- Pie chart for customer loyalty segmentation

10. Conclusion

Working on this project gave me a solid, hands-on understanding of how business intelligence tools can take raw data and turn it into something genuinely useful. By building a clean star schema data model, writing DAX measures, and putting together interactive dashboards, I was able to answer questions that any real automotive business would care about:

- Which products are actually selling well?
- Who are the most valuable customers and how do we keep them?
- Where and when should the business be focusing its energy?
- Is the service department meeting its targets and keeping customers satisfied?

The insights from this kind of analysis can genuinely help businesses make smarter decisions — whether that's fine-tuning marketing strategies, deciding where to invest in dealership infrastructure, building stronger relationships with loyal customers, or growing overall revenue. More than anything, this project showed me how much value there is in combining clean data, good design, and the right analytical tools.